

Seasoned Data Scientist and Machine Learning Engineer with 8+ years of end-to-end AI product delivery. Proven ability to translate challenges into scalable, production-grade solutions that integrate with medical systems, cloud platforms, and embedded applications. 15+ publications ranging from rare disease prediction, bias mitigation, to deep learning on EKG.

Experience

- **Senior Machine Learning Engineer**
 Visual Thought Labs

 - Developed ambient drug OCR in VA operating rooms, deployed on Nvidia Jetson at 77% lower hardware cost and 73% lower latency; built the SQL pipeline populating it, and HL7 system issuing results.

October 2025 - Present
 San Francisco, CA
- **Senior Data Scientist**
 UCSF

 - Built "Drifter," a data-drift monitoring tool used by 80 researchers across six hospitals; surfaced feature, label, and prediction drift that triggered timely ETL updates.
 - Directed a nation-wide matched-cohort study on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR = 1.67) and publishing findings in JAMA Network Open.

January 2022 - October 2025
 San Francisco, CA
- **Machine Learning Engineer**
 Syntegra

 - Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.

October 2021 - January 2022
 San Francisco, CA
- **Lead Data Scientist**
 LifeBell AI

 - Delivered early sepsis detection system using XGBoost, paired with explainer that highlighted abnormal labs. Deployed at Naval Medical Center San Diego.
 - Designed a Dynamic-Time-Warped Gaussian Mixture Model for phenotype discovery, surfacing sepsis-risk subgroups several hours before conventional detection.

December 2020 - October 2021
 Atlanta, GA (remote)
- **Data Scientist**
 UCSF

 - Developed deep-learning EKG classifier for pulmonary hypertension, achieving an AUC of 0.89 and early-diagnosis capability up to two years before conventional testing.
 - Led feasibility analyses that helped secure \$5M+ in research partnerships (J&J, Roche, Genentech).

August 2018 - December 2020
 San Francisco, CA
- **Embedded Systems Engineer**
 NASA Education & Public Outreach

 - Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.
 - Developed a STEM curriculum ("Learning By Making") that raised high-school science scores by 7.8% versus controls.

May 2014 - August 2018
 Rohnert Park, CA

Education

- **Stanford University** MS in Computational and Mathematical Engineering 2018
- **Sonoma State University** BS in Physics and Mathematics 2014

Skills

- **Languages** Python, SQL, Julia, C, C++, JavaScript, TypeScript
- **ML Frameworks** PyTorch, TensorFlow, Scikit-Learn, XGBoost, CVXPY, RAPIDS, JAX, TensorRT, Ollama, HuggingFace, StatsModels, lifelines
- **Infrastructure & Compute** Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker

Publications (Selected)

- C. Halabi, **H. Mills**, A. M. DiGiorgio, et al. "Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses". In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- A. S. Tang, K. P. Rankin, G. Ceroni, S. Miramontes, **H. Mills**, J. Roger, et al. "Leveraging electronic health records and knowledge networks for Alzheimer's disease prediction and sex-specific biological insights". In: *Nature Aging* 4.3 (2024), pp. 379–395.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, et al. "Electrocardiogram detection of pulmonary hypertension using deep learning". In: *Journal of cardiac failure* 29.7 (2023), pp. 1017–1028.